

MANAN RASTOGI

Applied AI Engineer | Computer Vision, LLM Evaluation, Full-Stack Systems

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EDUCATION

Vellore Institute of Technology (VIT), Vellore, India

2024 -- PRESENT

B.Tech in Electronics & Communication Engineering (Biomedical Specialization)

IIT Madras, Online, India

2025 -- PRESENT

BS in Data Science & Applications

TECHNICAL EXPERTISE

Languages & Frameworks: Python 3.12, TypeScript, React, Next.js, Node.js, Flask, OpenCV, SciPy, NumPy, TrackPy, Three.js.

Infrastructure & Tools: Git, Docker, Vercel, MongoDB Atlas, SQLite, CI/CD, LLM APIs (NVIDIA LLaMA 3.1), Tesseract OCR.

PROJECTS

WoundTrack AI: Applied Computer Vision Pipeline

2024 -- PRESENT

Python, OpenCV, Flask, SciPy, SQLite

[\[Live Demo\]](#) [\[Source\]](#)

- Built a CV pipeline tracking **441 cells** across **800+ frames** in **<10 min** on consumer hardware, achieving **28% higher precision** than manual ImageJ workflows.
- Implemented LoG blob detection + LAP gap-closing tracker with Kalman prediction for explainable cell localization and trajectory continuity through missed frames.
- Designed quantitative evaluation metrics (MSD exponents, velocity distributions, fractal analysis) validated across 5 experimental conditions.
- Built Flask + SQLite dashboard for batch processing, kinetic visualization, and automated report generation.

VITalize AI: Full-Stack Academic Platform

2025 -- PRESENT

Next.js 15, TypeScript, MongoDB, NVIDIA LLaMA 3.1, Tesseract OCR

[\[Live Demo\]](#)

- Built a unified academic platform indexing **3,000+** past exam papers with full-text search, subject-level filters, and bulk download.
- Integrated **NVIDIA LLaMA 3.1** API with Tesseract OCR fallback pipeline to extract text from scanned PDFs and generate unit-wise, mark-weighted AI solutions with KaTeX rendering.
- Engineered constraint-based **FFCS timetable solver** with clash elimination, faculty prioritization, gap minimization, and ICS calendar export.
- Deployed on **Vercel** with **MongoDB Atlas**; built Chrome extension for VTOP data sync.

Interactive 3D Engineering Portfolio

2024 -- PRESENT

TypeScript, React, Three.js (R3F), Framer Motion, Tailwind CSS

[\[Live Demo\]](#) [\[Source\]](#)

- Architected a React + TypeScript portfolio with lazy-loaded bundle splitting, sub-1s initial load, and responsive accessibility-first layout.
- Implemented WCAG accessibility: keyboard navigation, ARIA labels, skip-to-content, reduced-motion support, and focus management.
- Responsive layout with Framer Motion scroll-driven reveals and CSS-based reduced-motion fallback for accessibility.